ESTIMATING THE "TRUST PREMIUM" AND ADOPTION PROPENSITY IN INTERCITY SHARED MOBILITY: A DISCRETE CHOICE ANALYSIS

STRATEGIC CONTEXT AND PROBLEM IDENTIFICATION

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1. The Core Question

What are the primary econometric determinants of a consumer's willingness to pay a premium for intercity carpooling, and to what extent do physical safety apprehensions and demographic variables (specifically gender) dynamically moderate the probability of early app adoption?

2. Sector and Industry Description

The Indian intercity mobility sector serves hundreds of millions of travelers annually but operates in a highly polarized market. On one end is the unorganized, peer-to-peer shared mobility sector, which includes platforms like BlaBlaCar and regional Facebook carpool groups. This segment offers high affordability but suffers from severe reliability issues, such as last-minute cancellations, and notable physical safety risks. On the other end are private commercial fleet operators, including Uber Intercity and MakeMyTrip Cabs, which guarantee reliability but at a prohibitively high cost for frequent commuters. "Traxi" aims to bridge this gap by introducing a premium, asset-light carpooling model that monetizes peace of mind through strict corporate driver verification and zero-cancellation guarantees.

3. Why it Matters

In the highly competitive shared mobility space, customer acquisition costs are high, and platform loyalty is low. For the unit economics of Traxi to remain viable, specifically regarding their proposed ₹100 platform booking fee and the premium "Traxi Pink" tier, the company must econometrically validate whether the trust deficit translates into actual pricing power. If safety anxiety does not significantly increase a user's willingness to pay, the premium business model will fail against low-cost competitors.

4. Key Stakeholders

- **Platform Operators (Traxi):** These operators need this econometric data to optimize pricing algorithms, assess demand elasticity, and justify the financial costs of background verification.
- **Supply-Side Partners (Drivers):** Drivers require data to determine the optimal cost-sharing caps for their empty seats without pricing out consumers.
- **Regulatory Bodies and Policymakers:** These entities must assess how verified, private vehicle sharing economies can safely reduce highway congestion and carbon emissions without violating the Motor Vehicles Act.

5. Recent Trends and Key Issues

Following the pandemic, the global and Indian urban mobility landscapes have experienced a massive shift. There is a rising cost associated with private vehicle ownership, including insurance, fuel, and maintenance, which is driving a transition toward shared access over ownership, particularly among millennials and Generation Z. However, safety concerns and service unreliability remain the primary bottlenecks preventing the mass adoption of shared public transit and carpooling, particularly for female commuters. Platforms are increasingly relying on technology-enabled trust mechanisms, such as AI verification and corporate email validation, to overcome these behavioral barriers.

6. Supporting Credible Sources

- **NITI Aayog Report on Shared Mobility (2018):** This report highlights that while shared mobility has the potential to reduce vehicle kilometer demand in India by nearly 35 percent by 2035, issues regarding regulatory frameworks, passenger safety, and reliability must be aggressively addressed for the sector to scale beyond basic transit.
- **MDPI Sustainability Report on Shared Mobility (2024):** A recent structural equation modeling study focusing on Generation Z in India found that while economic benefits and environmental consciousness drive the intention to use shared mobility, perceived risks regarding safety and reliability act as a major indirect barrier. These perceived risks heavily moderate user attitudes and final adoption decisions.

DATA INTEGRITY AND EXPLORATORY INSIGHTS

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1. Data Processing and Cleaning

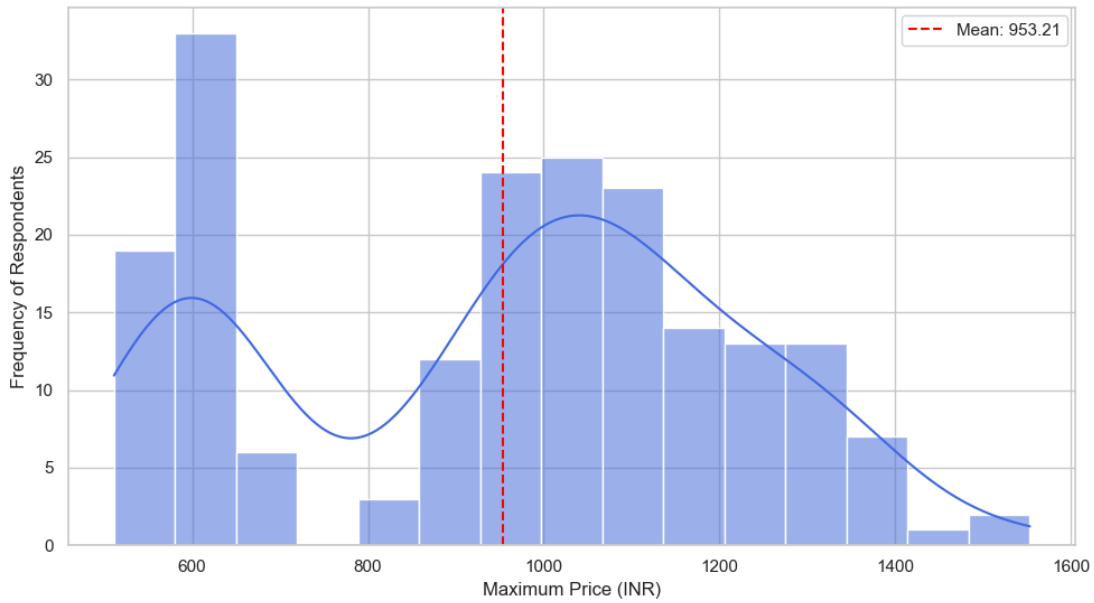
The empirical analysis relies on a primary survey dataset containing consumer responses regarding intercity mobility preferences. Initial data preprocessing involved inspecting the dataset for missing values and structural inconsistencies. No significant null values were detected in the primary target variables. To prepare the dataset for regression analysis, several categorical variables required transformation. The gender variable was converted into a binary dummy variable, where female respondents were assigned a value of 1 and male respondents a value of 0. Furthermore, categorical income strings were sanitized and encoded into separate dummy variables to prevent the dummy variable trap, establishing the lowest income bracket as the baseline reference. Finally, the dependent variable measuring download intent on a 1 to 7 scale was strictly transformed into a binary choice variable (where a score of exactly 7 equals 1, and all other scores equal 0). This strict transformation was deliberately executed to isolate absolute premium early adopters and prevent the econometric issue of complete separation, ensuring parameter identification in the subsequent Logit modeling.

2. Summary Statistics and Outlier Detection

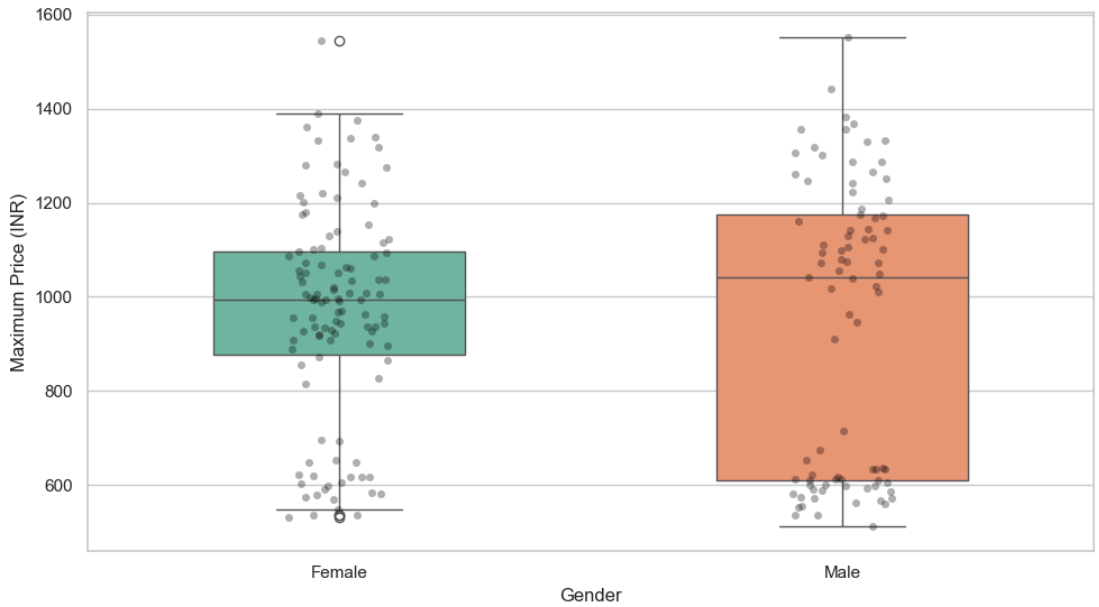
A preliminary review of the summary statistics reveals distinct behavioral patterns within the sample. The central tendency metrics for the maximum willingness to pay indicate a definitive baseline price tolerance among users. However, the standard deviation suggests significant variance across different demographic and psychographic segments. Potential outliers were identified in the willingness to pay variable, representing a small subset of users with an abnormally high price tolerance. Rather than trimming these observations and losing valuable data on premium customers, we will utilize a logarithmic transformation in our regression model. This standard econometric technique compresses the scale and mitigates the outsized influence of these extreme values without distorting the underlying behavioral trends.

3. Visualizations and Trend Identification

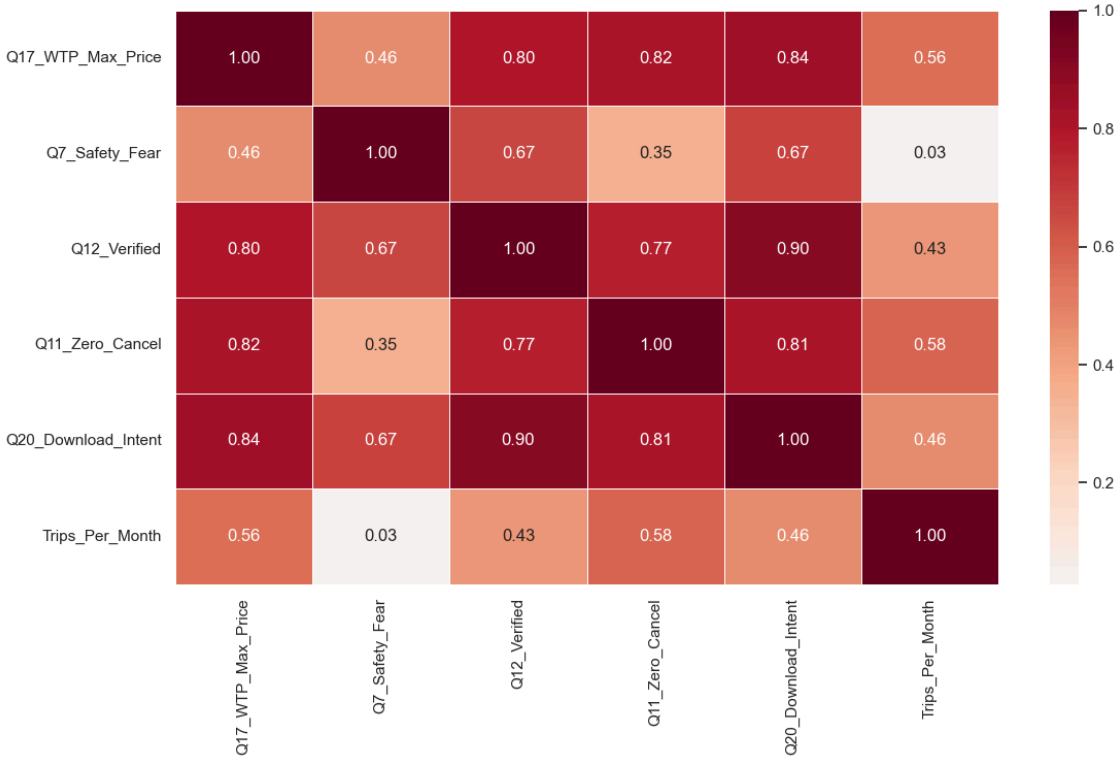
- **Distribution of Willingness to Pay (Histogram):** An analysis of the distribution of the maximum price users are willing to pay exhibits a noticeable right-skewed pattern. Most consumers cluster around a standard fare rate, while a long tail extends into premium pricing territory. This non-normal distribution visually justifies the upcoming use of a log-linear model, as taking the natural logarithm will normalize the dependent variable and allow us to estimate percentage changes.



- **Demographic Price Sensitivity (Boxplot):**
  A boxplot grouping the maximum willingness to pay by gender reveals a visible discrepancy in the median price tolerance. Female respondents appear to exhibit a higher baseline tolerance for premium pricing compared to male respondents. The interquartile range for female users is also tighter at a higher price point, suggesting a more consistent willingness to pay a premium. This visual evidence provides a strong empirical foundation for exploring demographic interaction terms in the modeling phase.



- **Psychometric and Behavioral Relationships (Correlation Heatmap):**
  A correlation matrix was generated to identify initial linear relationships between continuous and ordinal variables. The heatmap highlights strong positive associations between safety fear, the importance placed on driver verification, and the maximum willingness to pay. Additionally, there is a distinct positive correlation between the zero-cancellation guarantee and the intent to download the application. These correlations strongly suggest that the trust deficit serves as a quantifiable driver of consumer demand and pricing elasticity, rather than a mere psychological barrier.



HYPOTHESIS & VARIABLE SELECTION

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1. Justification of Dependent and Key Independent Variables

To rigorously test the viability of the Traxi business model, the econometric design requires variables that accurately capture both the financial tolerance and the behavioral intent of the consumer.

- **Dependent Variable 1: Maximum Willingness to Pay (Q17_WTP_Max_Price)** This continuous variable serves as the dependent variable for our primary Ordinary Least Squares regression. It quantifies the absolute monetary threshold a consumer is willing to cross for a shared intercity ride. Because it is a monetary value with a right-skewed distribution, we will apply a logarithmic transformation. This allows us to estimate the semi-elasticity of pricing, meaning we can measure the percentage increase in willingness to pay driven by a one-unit change in safety perception.
- **Dependent Variable 2: High Download Intent (High_Intent)** Derived from a seven-point Likert scale, this variable is transformed into a binary outcome where a value of 1 indicates high adoption intent and 0 indicates low intent. A discrete choice variable is fundamentally required to run a Logit model, allowing us to estimate the log-odds of a user actually adopting the platform based on the proposed safety features.
- **Key Independent Variables:**
    - **Safety Fear (Q7_Safety_Fear):** This continuous variable measures the psychological anxiety associated with unverified carpooling. It is the primary explanatory variable used to quantify the "Trust Deficit."
    - **Female Dummy (Female_Dummy):** This demographic dummy variable allows the model to control for baseline gender differences in pricing tolerance. More importantly, it serves as the foundation for an interaction term with "Safety Fear" to test the distinct price elasticity of female commuters.
    - **Corporate Verification Preference (Q12_Verified):** This variable measures how much the consumer values background checks. It represents the tangible product feature Traxi uses to mitigate safety fears.

2. Formulation of Clear and Testable Hypotheses

Based on the theoretical framework of the shared mobility market, we specify the following econometric hypotheses:

- **Hypothesis 1: The Safety Premium Effect**
    - **Null Hypothesis (H0):** A consumer's fear regarding physical safety has no statistically significant impact on their maximum willingness to pay for intercity carpooling.
    - **Alternative Hypothesis (HA):** There is a positive and statistically significant relationship between safety anxiety and willingness to pay. This implies that consumers actively assign a monetary premium to verified security.

- **Hypothesis 2: Gender Moderation and Price Elasticity**
    - **Null Hypothesis (H0):** The marginal effect of safety concerns on willingness to pay is identical across all genders.
    - **Alternative Hypothesis (HA):** Gender acts as a statistically significant moderator. Female users exhibit a higher marginal willingness to pay when safety fears increase compared to male users. Validating this hypothesis provides the economic justification for a premium, female-only "Traxi Pink" tier.
- **Hypothesis 3: Platform Adoption Propensity**
    - **Null Hypothesis (H0):** Placing a high importance on corporate driver verification does not significantly affect a consumer's probability of downloading the platform.
    - **Alternative Hypothesis (HA):** A higher preference for corporate-verified drivers significantly increases the log-odds of a consumer exhibiting high intent to adopt the application.

3. Potential Endogeneity Concerns and Omitted Variables

To ensure the integrity of our estimated coefficients, we must acknowledge potential violations of the core Ordinary Least Squares assumptions. The most pressing econometric concern in this model is Omitted Variable Bias, which leads to endogeneity.

Specifically, our dataset lacks a control variable for the "Average Distance of Usual Trip." Trip distance is naturally positively correlated with both a user's maximum willingness to pay and their general travel anxiety. By omitting this variable from the regression equation, its explanatory power is absorbed into the error term. Consequently, the error term becomes correlated with our key independent variable (Safety Fear). This violates the critical assumption that the covariance between the independent variables and the error term is zero. As a result, our model may suffer from an upward bias, artificially inflating the estimated coefficient for the safety premium.

Furthermore, there is a potential for simultaneity bias. High-income individuals might inherently possess a higher willingness to pay while simultaneously demanding higher safety standards due to their socioeconomic status. While our model mitigates this by including income dummy variables, unobserved wealth effects could still cause the error term to correlate with our regressors.

ECONOMETRIC MODEL SPECIFICATION AND RESULTS

To empirically test the formulated hypotheses regarding the trust premium and platform adoption propensity, we employ a two-equation econometric strategy. The analysis utilizes an Ordinary Least Squares (OLS) regression for continuous monetary valuation and a Maximum Likelihood Estimation (MLE) Logit model for discrete choice adoption behavior.

1. Model 1: The Trust Premium Model (Log-Linear OLS Regression)

To estimate the precise financial value consumers place on safety, we analyze the maximum willingness to pay. Because monetary variables inherently face lower bounds (zero) and typically exhibit a right-skewed distribution, applying a logarithmic transformation to the dependent variable is necessary to satisfy the OLS assumption of normally distributed residuals. Furthermore, a log-linear functional form allows the estimated coefficients to be interpreted as semi-elasticities, representing the percentage change in willingness to pay for a one-unit change in the independent variable.

We also introduce an interaction term between safety fear and the female demographic dummy. This allows the model to test whether the slope of pricing tolerance differs significantly between genders, relaxing the assumption of constant marginal effects across demographic groups. Finally, income brackets are included as categorical dummy variables to control for underlying purchasing power.

The Equation:

$$\begin{aligned}\ln(WTP_Max_Price_i) = \beta_0 + \beta_1 Safety_Fear_i + \beta_2 Female_Dummy_i + \\ \beta_3(Female_Dummy_i \times Safety_Fear_i) + \\ \sum_{k=1}^{3} \delta_k Income_Dummies_{ki} + u_i\end{aligned}$$

Variables Explained:

- **ln(WTP_Max_Price_i):** The natural logarithm of the maximum price user ***i*** is willing to pay.
- **β_1 :** Captures the baseline safety premium (the percentage increase in willingness to pay per unit increase in safety fear).
- **β_3 (Interaction Term):** Measures the differential impact of safety fear on pricing tolerance specifically for female users.
- **δ_k:** The coefficients for the income dummy variables, holding the lowest income bracket as the unobserved reference base to avoid perfect multicollinearity.

Model 1 Estimation Results:

(The following table presents the OLS estimation results extracted from the Python statsmodels library.)

```
MODEL 1: OLS REGRESSION (Estimating the Trust Premium)

Formula being used: Log_WTP ~ Q7_Safety_Fear * Female_Dummy + Income_15_25_LPA + Income_25_LPA_plus + Income_5_10_LPA + Income_lt__5_LPA

                            OLS Regression Results
==============================================================================
Dep. Variable:                Log_WTP   R-squared:                       0.746
Model:                            OLS   Adj. R-squared:                  0.736
Method:                 Least Squares   F-statistic:                     78.36
Date:                Wed, 08 Apr 2026   Prob (F-statistic):           3.19e-52
Time:                        17:11:44   Log-Likelihood:                 91.064
No. Observations:                 195   AIC:                            -166.1
Df Residuals:                     187   BIC:                            -139.9
Df Model:                           7
Covariance Type:            nonrobust
==============================================================================================
                                 coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------
Intercept                      6.4959      0.074     87.998      0.000       6.350       6.642
Q7_Safety_Fear                 0.0634      0.018      3.577      0.000       0.028       0.098
Female_Dummy                   0.0034      0.068      0.050      0.960      -0.131       0.138
Q7_Safety_Fear:Female_Dummy   -0.0050      0.019     -0.261      0.795      -0.043       0.033
Income_15_25_LPA               0.2399      0.047      5.076      0.000       0.147       0.333
Income_25_LPA_plus             0.3299      0.055      5.984      0.000       0.221       0.439
Income_5_10_LPA               -0.0991      0.048     -2.067      0.040      -0.194      -0.004
...
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
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2. Model 2: The Adoption Propensity Model (Logit Regression)

The second objective is to determine the drivers of platform adoption. The dependent variable is binary, where an absolute high intent to download the application equals 1, and all other responses equal 0. Applying standard OLS to a binary outcome creates a Linear Probability Model, which risks predicting probabilities outside the logical [0, 1] bounds and inherently violates the assumption of homoscedasticity. Therefore, we utilize a Logit Model, assuming a cumulative logistic distribution of the error term.

As noted in the data cleaning process, defining high intent too broadly resulted in perfect prediction (complete separation) by the independent variables. By restricting the dependent variable strictly to maximum intent, the Maximum Likelihood Estimator converges successfully.

The Equation:

$$L_i = \ln\left(\frac{P_i}{1-P_i}\right) = \alpha_0 + \alpha_1 Q12_Verified_i + \alpha_2 Q11_Zero_Cancel_i + \alpha_3 Trips_Per_Month_i + \epsilon_i$$

Variables Explained:

- **P_i:** The probability that user i exhibits maximum intent to adopt the platform.
- **L_i:** The Logit, representing the natural log of the odds ratio in favor of adoption.
- **α_1 and α_2:** The change in the log-odds of a user adopting the platform for every one-unit increase in how much they value verified drivers and zero cancellations, respectively.
- **α_3:** A control variable accounting for the frequency of the user's intercity travel.

Model 2 Estimation Results:

(The following table presents the Logit MLE results extracted from the Python statsmodels library.)

```
MODEL 2: LOGIT REGRESSION (Estimating Adoption Propensity)

Optimization terminated successfully.
         Current function value: 0.526446
         Iterations 7
                           Logit Regression Results
==============================================================================
Dep. Variable:          High_Intent_7   No. Observations:                  195
Model:                          Logit   Df Residuals:                      191
Method:                           MLE   Df Model:                            3
Date:                Wed, 08 Apr 2026   Pseudo R-squ.:                  0.2006
Time:                        17:11:44   Log-Likelihood:                -102.66
converged:                       True   LL-Null:                       -128.42
Covariance Type:            nonrobust   LLR p-value:                 3.793e-11
=====================================================================================
                      coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------------
const              -5.6825      1.061     -5.358      0.000      -7.761      -3.604
Q12_Verified        0.4252      0.182      2.332      0.020       0.068       0.782
Q11_Zero_Cancel     0.4704      0.189      2.489      0.013       0.100       0.841
Trips_Per_Month     0.0620      0.154      0.402      0.688      -0.240       0.364
=====================================================================================
```

PREDICTIVE PERFORMANCE & HYPOTHESIS VALIDATION

1. Overall Model Fit and Diagnostics

Before interpreting individual coefficients, we must establish the explanatory power and validity of both econometric models. The table below summarizes the core diagnostic metrics.

Metric Category	Model 1: OLS (Trust Premium)	Model 2: Logit (Adoption Propensity)
Primary R-Squared	0.746 (Adjusted: 0.736)	0.2006 (McFadden's Pseudo)
Global Significance	F-statistic: 78.36	LLR p-value: 3.793e-11
Model p-value	3.19e-52 (Highly Significant)	Highly Significant
Key Assumption Check	Durbin-Watson: 1.898 (No Autocorrelation)	Successfully resolved complete separation

The Ordinary Least Squares log-linear regression explains approximately 74.6 percent of the variance in a consumer's maximum willingness to pay. The F-statistic confirms that our independent variables jointly exert a significant influence on pricing tolerance. For the qualitative choice model, the Logit regression achieved a Pseudo R-squared of 0.2006. In qualitative response models, a Pseudo R-squared between 0.2 and 0.4 indicates an excellent fit. Both models are highly statistically significant compared to a null baseline. Furthermore, the Durbin-Watson statistic of 1.898 confirms there is no severe first-order autocorrelation present in our OLS residuals.

2. Hypothesis Testing Results

The core of this analysis relies on testing our three formulated hypotheses. The empirical outcomes are summarized in the following table.

Hypothesis	Key Independent Variable	Coefficient	P-Value	Statistical Conclusion
H1: Safety Premium Effect	Safety Fear	0.0634	0	Reject Null Hypothesis
H2: Gender Moderation	Safety Fear * Female Dummy	-0.005	0.795	Fail to Reject Null
H3: Adoption Propensity	Driver Verification	0.4252	0.02	Reject Null Hypothesis
H3: Adoption Propensity	Zero Cancellations	0.4704	0.013	Reject Null Hypothesis

The Safety Premium Effect (H1): The regression output strongly supports our first hypothesis. The coefficient for safety fear is highly statistically significant. Because our dependent variable is log-transformed, we interpret this as a semi-elasticity. Holding all else constant, a one-unit increase in a user's self-reported safety fear results in an approximate 6.34 percent increase in their maximum willingness to pay. This confirms the trust premium is a quantifiable phenomenon.

- **Gender Moderation and Interaction (H2):** The interaction term between safety fear and the female demographic dummy yielded a p-value far above the standard 0.05 threshold. Similarly, the standalone baseline female dummy was statistically insignificant ($p = 0.960$). Econometrically, this indicates that while safety anxiety heavily drives pricing tolerance, the rate at which it does so is statistically uniform across both men and women in this sample.
- **Platform Adoption Propensity (H3):** Valuing corporate verification and zero cancellations successfully drives the probability of platform adoption. For every one-unit increase in the importance a user places on verification, the log-odds of them being a high-intent adopter increases by 0.4252. Similarly, the zero cancellation variable is highly significant. We conclude that these specific platform features are primary drivers of user acquisition.

3. Control Variables and Income Dynamics

Beyond our primary explanatory variables, the models included controls for purchasing power and travel frequency to isolate the true effects of safety and reliability.

Control Variable	Associated Model	Coefficient	P-Value	Economic Interpretation
Income: >25 LPA	Model 1 (OLS)	0.3299	0	Willing to pay 32.99% more than baseline
Income: <5 LPA	Model 1 (OLS)	-0.2287	0	Willing to pay 22.87% less than baseline
Trips Per Month	Model 2 (Logit)	0.062	0.688	Travel frequency does not drive adoption

The income dummy variables behaved precisely as expected according to standard economic theory. Using the 10 to 15 LPA bracket as the unobserved reference base, we observe clear structural changes in spending limits among consumers. Conversely, the control variable measuring the frequency of travel yielded an insignificant p-value. This implies that the adoption of a premium platform like Traxi is driven purely by the desire for safety and reliability, rather than by how frequently a user actually travels.

SYNTHESIS OF FINDINGS & STRATEGIC RECOMMENDATION

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1. Interpretation of Real-World Implications

The empirical findings of this study directly validate Traxi's foundational business hypothesis: consumers in the intercity mobility market do not view safety merely as a baseline expectation; they treat it as a premium economic commodity. The econometric models provide actionable insights for platform strategy:

- **The Viability of the Trust Premium:** The OLS log-linear model proved that a one-unit increase in safety anxiety translates to a 6.34 percent increase in the absolute monetary threshold a user is willing to cross. This confirms that Traxi can successfully implement a premium pricing strategy compared to unverified competitors, as consumers actively assign financial value to physical security.
- **Drivers of Platform Acquisition:** The discrete choice Logit model confirmed that corporate verification and zero-cancellation guarantees are the defining econometric drivers of early app adoption. Marketing efforts should focus exclusively on these reliability metrics rather than frequency-based discounts, as travel frequency was proven statistically insignificant in driving high adoption intent.
- **Reevaluating the Gender-Based Strategy:** The interaction term testing gender price elasticity failed to achieve statistical significance. Econometrically, this indicates that while both men and women heavily value safety, female commuters are not statistically *more* tolerant of price hikes than male commuters when safety is concerned. Therefore, while a premium "Traxi Pink" tier is operationally viable for safety reasons, aggressively overpricing it solely based on demographic assumptions could lead to demand destruction.

2. Limitations and Endogeneity Concerns

While the estimated models are highly robust, they are constrained by potential endogeneity, primarily driven by Omitted Variable Bias (OVB).

- **Omitted Variables:** The dataset lacks control variables for the "Average Distance of Usual Trip" and a consumer's "Historical Experience with Ride-hailing Crime." Trip distance is naturally positively correlated with both a user's maximum willingness to pay and their general travel anxiety.
- **Impact on Estimation:** By omitting these variables from the regression equation, their explanatory power is absorbed into the error term (u_i). Consequently, the error term becomes correlated with our key independent variable (Safety Fear). This violates the critical zero-conditional-mean assumption of OLS, potentially leading to an upward bias in our estimated trust premium coefficient.

3. Scope for Further Research

To refine the pricing model and mitigate the identified econometric limitations, future iterations of this research should adopt the following approaches:

- **Instrumental Variables (IV):** Future studies should employ an Instrumental Variable approach or a Two-Stage Least Squares (2SLS) regression to isolate the exogenous variation in safety fear, thereby neutralizing the omitted variable bias.
- **Longitudinal Panel Data:** The current analysis relies on self-reported survey intent, which is susceptible to hypothetical bias. Transitioning to panel data that tracks actual user conversion rates and real financial transactions on the platform will allow for the estimation of dynamic, real-world price elasticity over time.